

TD2 process

Exercise 1

- Write a program that creates a child. The father should display "I am the father" and the child should display "I am the child".
- Write a program that creates 2 children :child 1 and child 2. Father should display "I am the father" and child 1 should display "I am child 1", and child 2 should display "I am the child 2".
- C. Write a program that creates 5 children using a for loop. It will be noted that so that the child does not itself create several children, the loop must be interrupted by a break in the child. We can use a global variable

Exercise 2

Write a new version of the "cp" command that allows you to make multiple copies of a file in parallel. The syntax of the command must be of the form:

cp1 source_file_dest1_file2dest_file3

The parent process creates as many processes as there are copies to make, each child process makes a copy (son1 copies file_srouce under file_dest1; son2 copies source_file under file_dest2 etc ...). The parent process waits for the termination of its children one by one and displays at each termination a message that the copy of such file was successful (or an error message in the case of a copy failure). Use the "system (char * command)" function to perform the copy operation on child processes.

Exercise 3

Write a program that creates two children called children 1 and children 2. Each child must wait for a number of random seconds between 1 and 10, using the sleep function.

The program waits for the longest child to finish and displays the total time. We can use the time function of the time.h library, which returns the number of seconds since January 1, 1970 at 0h (in universal time).

Exercise 4

Write a program that performs the sum of the elements of the rows of a matrix in parallel. Initially the parent process reads the elements of the matrix in an array, it then starts to create the processes and assign to each the calculation of the sum for a row of the matrix (we create as many processes as there are rows) . After the creation of the processes, the parent process waits for the termination of its children to collect their calculation result.

Exercise 5

What does each of the following programs display (Justify):

Programme A)

```
#include<stdio.h>
#include<signal.h>
void doit() {
    fork();
    fork();
    printf("Bravo !\n");
    fork();
}
int main() {
    doit();
    printf("Bravo !\n");
```

12 Bravo !

```
exit(0);
}
```

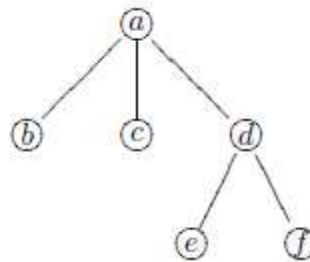
Programme B)

```
#include<stdio.h>
#include<signal.h>
int main() {
    pid_t pid;
    int x = 1;
    pid = fork();
    if (pid == 0) {
        printf("Dans fils:x=%d\n", ++x);
        exit(0);
    }
    printf("Dans pere : x=%d\n",x);
    exit(0);
}
```

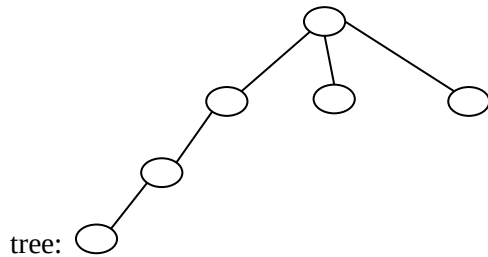
Exercise 6

- Write a program in C, which generates the following process

```
if(fork()) {
    fork() && fork();
} else {
    fork() && fork();
}
```



a)

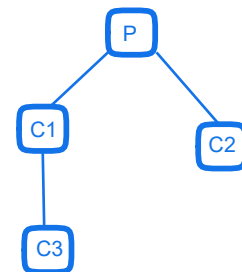


```
if(fork()) {
    fork() && fork();
} else {
    fork() || fork();
}
```

b)

- Draw the family tree of the processes generated by the following program:ex1.c

```
1 : #include <unistd.h>
2 :
3 : int main(void)
4 : {
5 :     fork() && (fork() || fork());
6 :     return 0;
7 : }
```



Exercise 7

Write a C program to run the command passed as an argument.

Exercise 8

Write a C program equivalent to the following shell command:

- who & ps & ls -l
- who; ps; ls -l